Some computations *On maximizing the number of heads...*

16. April 2026

Rewriting the basic recursion

Let us rewrite (1) in the manuscript by using $w_{n,p} := n - v_{n,p}$, i.e. $w_{n,p}$ is the expected number of tails set aside during the experiment. This gives

$$\begin{aligned} w_{n,p} &= n(p^n + (1-p)^n) - np^n - v_{n-1,p}(1-p)^n + \sum_{j=1}^{n-1} \binom{n}{j} p^j (1-p)^{n-j} \Big(\underbrace{n - \max_{1\le i\le j}(v_{n-i,p} + i)}_{=\min_{1\le i\le j} w_{n-i,p}} \Big) \\ &= (w_{n-1,p}+1)(1-p)^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \Big(\min_{j\le i\le n-1} w_{i,p} \Big). \end{aligned}$$

At least for me, this is a bit easier to grasp than (1) in the manuscript.

Some notes on strategy A

Recall that strategy A consists of only setting aside one coin, unless all are heads. In the latter case, set aside all coins. (Also, recall that strategy A is optimal iff $v_{n,p} \ge v_{n-1,p} + 1$ for all n.) This is the case iff

$$w_{n,p} = n - v_{n,p} \le n - v_{n-1,p} - 1 = w_{n-1,p},$$

i.e. the map $i \mapsto w_{i,p}$ is decreasing, and we find $\min_{j\le i\le n-1} w_{i,p} = w_{n-1,p}$.) For strategy A, we have

$$w_{n,p} = (1-p)^n + (1-p^n) w_{n-1,p}$$

and this recursion is readily solved by

$$w_{n,p} = \sum_{k=1}^{n} (1-p)^k (1-p^{k+1}) \cdots (1-p^n).$$

In fact, this formula is not surprising. We can write

$$\begin{aligned} w_{n,p} &= \mathbb{E}\Big[\sum_{k=1}^{n} 1_{\text{set aside a tail when there are } k \text{ coins left}} \Big] \\ &= \sum_{k=1}^{n} \mathbb{P}(\text{set aside a tail when there are } k \text{ coins left}), \end{aligned}$$

and we set aside a tail when there are k coins left iff there were less than ℓ heads when having $\ell = k+1, ..., n$ coinds left (otherwise we would not have come to the situation of having k coins left), and there are all tails when throwing k coins.

Changing the game a bit

Let us change the game a bit: you only win if all coins come up heads, otherwise you lose. What is the best strategy for winning (depending on p)?

In this case, we get the Bellman equation for $w_{n,p} = \max_s \mathbb{P}_s(\text{winning with } n \text{ coins left})$, where $\mathbb{P}_s$ is the law when playing strategy s with n coins. We get

$$w_{n,p} = p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \max_{j \le i \le n-1} w_{i,p} \tag{1}$$

eq:wn

Let us start simple and compute the first few terms. Clearly, $w_{1,p} = p$. Then,

$$\begin{aligned} w_{2,p} &= p^2 + 2p^2(1-p) = p^2(3-2p), \\ w_{3,p} &= p^3 + 3p(1-p)((1-p)w_{2,p} + p(w_{1,p} \vee w_{2,p})). \end{aligned}$$

Here, note that $w_{1,p} > w_{2,p}$ iff $2p^2 - 3p + 1 = (1-2p)(1-p) > 0$, i.e. $p < 1/2$. So, in this case, if we roll three coins and two come up heads, it is best to set aside both of them, whereas for $p > 1/2$, it is better to set aside only one.

We can show the following:

l1 **Lemma 0.1.** *Let $p > 1/2$. Then, for all $n = 2, 3, ...,$*

$$w_{n,p} > w_{n-1,p} \quad \text{and} \quad w_{n-1,p} < \frac{p^n}{(1-p)^n + p^n}.$$

Corollary 0.2. *Let $p > 1/2$. Then $n \mapsto w_{n,p}$ is increasing.*

Proof of Lemma 0.1. First, note that

$$n \mapsto \frac{p^n}{(1-p)^n + p^n} = \frac{1}{1 + ((1-p)/p)^n} \text{ is increasing} \tag{2}$$

eq:in1

since $(1-p)/p < 1$.

We proceed by induction. For $n = 2$, we have already shown $w_{2,p} > w_{1,p}$ as well as $w_{1,p} = p = \frac{p}{(1-p)+p} < \frac{p^2}{(1-p)^2+p^2}$ by (2). Next, assume that the assertion holds for $w_{2,p}, ..., w_{n-1,p}$. Then,

$$\begin{aligned} w_{n,p} &= p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j w_{n-1,p} \\ &= p^n + (1 - p^n - (1-p)^n) w_{n-1,p} = w_{n-1,p} + p^n(1 - w_{n-1,p}) - (1-p)^n w_{n-1,p}. \end{aligned}$$

Clearly, we have $w_{n,p} > w_{n-1,p}$ iff

$$\frac{(1-p)^n}{p^n} < \frac{1 - w_{n-1,p}}{w_{n-1,p}} = \frac{1}{w_{n-1,p}} - 1, \quad \text{i.e.} \quad w_{n-1,p} < \frac{p^n}{(1-p)^n + p^n},$$

which holds by induction hypothesis. Last, we find that

$$\begin{aligned} w_{n,p} &= p^n + (1 - p^n - (1-p)^n) w_{n-1,p} \\ &< p^n \Big(1 + \frac{1 - p^n - (1-p)^n}{(1-p)^n + p^n}\Big) = \frac{p^n}{(1-p)^n + p^n} < \frac{p^{n+1}}{(1-p)^{n+1} + p^{n+1}} \end{aligned}$$

by (2) and we are done. □

No convergence to 0

If p is small, one might be tempted to conjecture that $w_{n,p} \xrightarrow{n\to\infty} 0$, i.e. for a large number of coins, there is no chance for a win. However, if we choose strategy A, we win iff, independently for all $n = 1, 2, ...$, we never have only tails. This gives

$$\begin{aligned}\lim_{n\to\infty} w_{n,p} &\geq \prod_{k=1}^{\infty}(1-(1-p)^k) = \exp\big(\sum_{k=1}^{\infty}\log(1-(1-p)^k)\big)\\ &\geq \exp\big(-\sum_{k=1}^{\infty}(1-p)^k\big) = \exp(-(1-p)/p) > 0.\end{aligned}$$

Two concrete strategies

Let us study two concrete strategies:

(A) Always put aside one coin (with head) if there is one, unless all are head. In this case, put aside all (and win).

(B) Always put aside as many coins as there are heads.

The probability for a win with strategy (A) is

$$\begin{aligned}p_n &= p^n + (1-(1-p)^n - p^n)p_{n-1},\\ p_n &= \sum_{k=1}^{n} p^k(1-(1-p)^{k+1} - p^{k+1})\cdots(1-(1-p)^n - p^n).\end{aligned}$$

As $n \to \infty$, this gives approximately

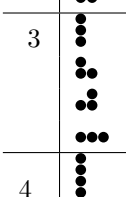
$$p_n = \sum_{k=1}^{\infty} p^k \prod_{\ell=k+1}^{\infty}(1-(1-p)^\ell - p^\ell)$$

For (B), we note the following: Every coin has an independent geo(p)-distribution for the first time it comes uo heads. Let $T_1, T_2, ...$ be these independent random variables. Then, we can describe the probability to win as

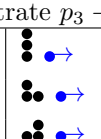
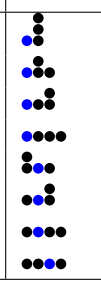
$$p_n = \mathbf{P}(\text{there is } t = 1, 2, ... \text{ such that } T_1, ..., T_n \leq t \text{ and for all } s \leq t \text{ there is } i \text{ with } T_i = s).$$

$$\begin{aligned}p_n &= \sum_{t=1}^{n}\sum_{\substack{k_1,\dots,k_t\geq 1\\ k_1+\cdots+k_t=n}}\binom{n}{k_1\cdots k_t}\prod_{s=1}^{t}\Big((1-p)^{s-1}p\Big)^{k_s}\\ &= \Big(\frac{p}{1-p}\Big)^n\sum_{t=1}^{n}\sum_{\substack{k_1,\dots,k_t\geq 1\\ k_1+\cdots+k_t=n}}\binom{n}{k_1\cdots k_t}(1-p)^{\sum_{s=1}^{t} sk_s}\end{aligned}$$

Let us illustrate this formula:

n	case	probability
1		p
2		p^2
		$2p^2(1-p)$
3		p^3
		$3p^3(1-p)$
		$3p^3(1-p)^2$
		$6p^3(1-p)^3$
4		p^4
		$4p^4(1-p)$
		$6p^4(1-p)^2$
		$4p^4(1-p)^3$
		$12p^4(1-p)^3$
		$12p^4(1-p)^4$
		$12p^4(1-p)^5$
		$24p^4(1-p)^6$

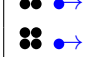
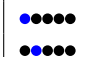
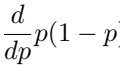
Let us illustrate $p_3 - p_4$:

$W_3 \setminus W_4$		$p^3(1-p)^2$
		$3p^3(1-p)^4$
		$3p^3(1-p)^5$
		$6p^3(1-p)^7$
$W_4 \setminus W_3$		$p^4(1-p)^3$
		$3p^4(1-p)^4$
		$3p^4(1-p)^5$
		$6p^4(1-p)^6$
		$3p^4(1-p)^3$
		$3p^4(1-p)^5$
		$6p^4(1-p)^6$
		$6p^4(1-p)^6$

First:

$$
\begin{aligned}
p^3(1-p)^2 &\geq 4p^4(1-p)^3,\\
3p^3(1-p)^5 &\geq 12p^4(1-p)^6,\\
3p^3(1-p)^4 + 6p^3(1-p)^7 - 3p^4(1-p)^4 - 6p^4(1-p)^5 - 6p^4(1-p)^6&\\
\geq 6p^3(1-p)^6(1-2p) + 3p^3(1-p)^5 - 6p^4(1-p)^5 &\geq 0
\end{aligned}
$$

Let us illustrate $p_4 - p_5$:

$W_4 \setminus W_5$		$p^4(1-p)^2$
		$4p^4(1-p)^4$
		$6p^4(1-p)^5$
		$4p^4(1-p)^6$
		$12p^4(1-p)^7$
		$12p^4(1-p)^8$
		$12p^4(1-p)^9$
		$24p^4(1-p)^{11}$
$W_5 \setminus W_4$		$p^5(1-p)^4$
		$4p^5(1-p)^5$
		$4p^5(1-p)^3$
		$6p^5(1-p)^6$
		$6p^5(1-p)^5$
		$4p^5(1-p)^7$
		$6p^5(1-p)^7$
		$12p^5(1-p)^7$
		$12p^5(1-p)^6$
		$12p^5(1-p)^6$
		$12p^5(1-p)^8$
		$12p^5(1-p)^8$
		$12p^5(1-p)^7$
		$12p^5(1-p)^9$
		$12p^5(1-p)^9$
		$12p^5(1-p)^9$
		$24p^5(1-p)^{10}$
		$24p^5(1-p)^{10}$
		$24p^5(1-p)^{10}$
		$24p^5(1-p)^{10}$

$$\frac{d}{dp}p(1-p)^k = (1-p)^k - kp(1-p)^{k-1} = (1-p)^{k-1}(1-p(k+1)),$$

so

$$\frac{(k+1)^{k+1}}{k^k}p(1-p)^k \leq 1.$$

n case probability

With $p_0 = 1$,

$$
\begin{aligned}
p_n &= \sum_{k=0}^{n-1} \binom{n}{k} (1-p)^k p^{n-k} p_k. \\
\sum_{n=1}^{\infty} p_n s^n &= \sum_{k=0}^{\infty} \sum_{n=k+1}^{\infty} \binom{n}{k} (1-p)^k p^{n-k} p_k s^n \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} ((1-p)s)^k p_k \sum_{n=k+1}^{\infty} n \cdots (n-k+1)(ps)^{n-k} \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} ((1-p)s)^k p_k \frac{d^k}{dx^k} \sum_{n=1}^{\infty} x^n \Big|_{x=ps} \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} ((1-p)s)^k p_k \frac{d^k}{dx^k} \Big(\frac{1}{1-x} - 1 \Big) \Big|_{x=ps} \\
&= \frac{ps}{1-ps} + \sum_{k=1}^{\infty} \frac{1}{k!} ((1-p)s)^k p_k (-1)^k k! (1-ps)^{-(k+1)} \\
&= \frac{ps}{1-ps} - \frac{1}{1-ps} \sum_{k=1}^{\infty} p_k \Big(\frac{-(1-p)s}{1-ps} \Big)^k
\end{aligned}
$$

Note that

$$
\begin{aligned}
\mathbf{P}(\max_{i=1,\dots,n} T_i \le t - \log_{1-p} n) &= (1 - (1-p)^{t - \log_{1-p} n})^n \approx \exp(-(1-p)^t), \\
\mathbf{P}(\max_{i=1,\dots,n} T_i = \lfloor t - \log_{1-p} n \rfloor) &\approx \exp(-(1-p)^t) - \exp(-(1-p)^{t-1}), \\
\mathbf{P}\Big(\bigcup_{i=1}^{n} \{T_i = a - \log_{1-p} n\} \Big) &= 1 - \prod_{i=1}^{n} \Big(1 - (1-p)^{a - \log_{1-p} n - 1} p \Big) = 1 - \Big(1 - (1-p)^{a-1} p/n \Big)^n \\
&\approx 1 - \exp\Big(-(1-p)^{a-1} p \Big) \\
p_n &\approx \sum_{t=-\infty}^{\infty} \big(\exp(-(1-p)^t) - \exp(-(1-p)^{t-1}) \big) \prod_{a=-\infty}^{t-1} \Big(1 - \exp\Big(-(1-p)^{a-1} p \Big) \Big)
\end{aligned}
$$

$X \sim \exp(1)$, so

$$
\mathbf{P}(ts \le X \le (t+1)s) = e^{-ts} - e^{-(t+1)s} = e^{-ts}(1 - e^{-s}).
$$

For $Z := \max_{i=1,\dots,n} X_i$,

$$
\begin{aligned}
\mathbf{P}(Z \in dz) &= \frac{d}{dz} (1 - e^{-z})^n = n e^{-z} (1 - e^{-z})^{n-1}, \\
\mathbf{P}(Z - \lfloor Z \rfloor < x) &= \sum_{k=0}^{\infty} \int_k^{k+x} n e^{-z} (1 - e^{-z})^{n-1} dz = \sum_{k=0}^{\infty} (1 - e^{-z})^n \Big|_{z=k}^{z=k+x} \\
&= \sum_{k=0}^{\infty} (1 - e^{-(k+x)})^n - (1 - e^{-k})^n \\
&= \sum_{k=0}^{\infty} (1 + e^{-x})^n (1 - e^{-k})^n + -(1 - e^{-k})^n
\end{aligned}
$$

For large n,

$$\begin{aligned} p_n &:= \sum_t \mathbf{P}(\text{success}, \max_{i=1,\dots,n} T_i = t) \\ &\approx \sum_t \mathbf{P}(\bigcap_k \bigcup_i 1_{T_i = t-k}, \max_{i=1,\dots,n} T_i = t) \end{aligned}$$

Heuristics with $p = 1 - e^{-s}$, so $1 - p = e^{-s}$

$$p_\infty \approx \prod_{k=1}^{\infty} (1 - e^{-ks}) = \prod_{k=1}^{\infty} (1 - (1-p)^k)$$

$$\begin{aligned} &a_1 = 1, \\ &a_2 = p + (1-p)p, \\ &a_3 = p^2 + p^2(1-p) + (1-p)^2 a_2 p_n \qquad \approx p^2 + 2p(1-p)p + (1-p)^2 \end{aligned}$$

Here is another probabilsitic model capturing the model: Take $X_1, \dots, X_n \sim \exp(1)$ independently, and independent Poisson processes $P_1, \dots, P_n$ with rate $\lambda > 0$ with $p = 1/(1+\lambda)$ or $\lambda = (1-p)/p$. Then, by construction, $T_i = |\{t \le X_i : t \in P_i\}| \sim geo(p)$. Rather, let us assume that there is a single Poisson process P, which makes the resulting geometric RVs dependent. Then we want to compute the probability that between any two clicks of P there is at least one i such that X_i runs off. This can be computed via

$$\prod_{k=1}^{n} \frac{k}{k+\lambda}.$$

Let $p_{n,t}$ be the probability of a win when starting with n coins after exactly t coin tosses. Then,

$$p_{n+1,t} = p_{n,t}(1 - (1-p)^t) + p_{n,t-1}(1-p)^{t-1} p$$

So, with $a_n(s) = \sum_t p_{n,t} s^t$,

$$a_{n+1}(s) = \sum_t p_{n+1,t} s^t = a_n(s) - a_n(s(1-p)) + spa_n(s(1-p)) = a_n(s) - a_n(s(1-p))(1-sp)$$

$$P(S_{n+1} | \mathcal{F}_n) = 1_{S_n}(1 - (1-p)^{T_n+1})$$

$$\mathbf{E}[A_n 1_{S_{n+1}} | \mathcal{F}_n] = A_n 1_{S_n}(1 - (1-p)^{T_n+1}) = A_{n-1} 1_{S_n}$$

So, $A_n = \prod_{k=1}^{n} (1 - (1-p)^{T_k+1})$

For the case $p < 1/2$ we have the following:

Lemma 0.3. *Let $p < 1/2$. If $w_{n-1,p} = \max_{k=1,\dots,n-1} w_{k,p}$. Then, $w_{n,p} < w_{n-1,p}$.*

Proof. First, note that (since $p < 1/2$),

$$n \mapsto \frac{p^n}{(1-p)^n + p^n} = \frac{1}{1 + ((1-p)/p)^n} \text{ is decreasing,}$$

in particular $\frac{p^n}{(1-p)^n+p^n} < p = w_{1,p}$. From here on, we copy the calculations from Lemma 0.1: We compute

$$\begin{aligned} w_{n,p} &= p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j w_{n-1,p} \\ &= p^n + (1 - p^n - (1-p)^n) w_{n-1,p} = w_{n-1,p} + p^n(1 - w_{n-1,p}) - (1-p)^n w_{n-1,p}. \end{aligned}$$

Clearly, we have $w_{n,p} < w_{n-1,p}$ iff

$$\frac{(1-p)^n}{p^n} > \frac{1 - w_{n-1,p}}{w_{n-1,p}} = \frac{1}{w_{n-1,p}} - 1, \quad \text{i.e.} \quad w_{n-1,p} > \frac{p^n}{(1-p)^n + p^n}.$$

The latter holds since $w_{n-1,p} \geq w_{1,p} = p > \frac{p^n}{(1-p)^n+p^n}$. $\square$

Let us study the case $p < 1/2$. Setting $v_0 := 1$, note that (1) changes to

$$\begin{aligned}
v_{n,p} &= \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j v_j \\
&= \sum_{j=0}^{n-1} \Big(\binom{n-1}{j} + \binom{n-1}{j-1} \Big) p^{n-j}(1-p)^j v_j \\
&= p v_{n-1}(1 + (1-p)^{n-1}) + \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{j+1}(v_j + v_{j+1} - v_j) \\
&= v_{n-1} + p(1-p)^{n-1} v_{n-1} + (1-p) \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^j (v_{j+1} - v_j) \\
&= v_{n-1} + p(1-p)^{n-1} v_{n-1} - (1-p)^n (v_n - v_{n-1}) + (1-p) \sum_{j=0}^{n-1} \binom{n-1}{j} p^{n-1-j}(1-p)^j (v_{j+1} - v_j)
\end{aligned} \tag{3}$$

eq:wn1

So, we find that

$$v_n < v_{n-1} \quad \Longleftrightarrow \quad p(1-p)^{n-1} v_{n-1} < (1-p) \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^j (v_j - v_{j+1})$$

With

$$\begin{aligned}
&\sum_{j=0}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j (v_{j+1} - v_j) \\
&\quad = \sum_{j=1}^{n} \Big(\binom{n-1}{j-1} + \binom{n-1}{j-2} \Big) p^{n-j+1}(1-p)^{j-1} v_j - \sum_{j=0}^{n-1} \Big(\binom{n-1}{j-1} + \binom{n-1}{j} \Big) p^{n-j}(1-p)^j v_j \\
&\quad = (1-p)^n v_n + (2p-1) \sum_{j=1}^{n} \binom{n-1}{j-1} p^{n-j}(1-p)^{j-1} v_j \\
&\quad + \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-j-1}(1-p)^{j+1} v_{j+2} - \sum_{j=0}^{n-1} \binom{n-1}{j} p^{n-j}(1-p)^j v_j
\end{aligned}$$

Note that

$$\begin{aligned}
\sum_{j=i+1}^{n-1} \binom{n}{j}\binom{j}{i}(1-p)^{j-i} &= \frac{n!}{i!(n-i)!} \sum_{j=i+1}^{n-1} \binom{n-i}{j-i}(1-p)^{j-i} \\
&= \binom{n}{i} \sum_{j=1}^{n-i-1} \binom{n-i}{j}(1-p)^j \\
&= \binom{n}{i} \big((2-p)^{n-i} - 1 - (1-p)^{n-i} \big)
\end{aligned}$$

Pluggin in this result for $v_{j+1} - v_j$, we obtain

$$\begin{aligned}
v_{n,p} - v_{n-1} &= p(1-p)^{n-1} v_{n-1} + (1-p) \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j} (1-p)^j \\
&\qquad \cdot \Big(p(1-p)^j v_j + (1-p) \sum_{i=0}^{j-1} \binom{j}{i} p^{j-i} (1-p)^i (v_{i+1} - v_i) \Big) \\
&= p(1-p)^{n-1} v_{n-1} + (1-p) \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j} (1-p)^j p (1-p)^j v_j \qquad (4) \quad \boxed{\texttt{eq:wn1}} \\
&+ (1-p)^2 \sum_{i=0}^{n-3} p^{n-1-i} (1-p)^i (v_{i+1} - v_i) \underbrace{\sum_{j=i+1}^{n-2} \binom{n-1}{j} \binom{j}{i} (1-p)^j}_{= \frac{(n-1)!}{i!(n-1-i)!} \sum_{j=i+1}^{n-2} \binom{n-1-i}{j-i} (1-p)^j}
\end{aligned}$$

Let (with $x_n = v_{n+1} - v_n$),

$$x_n = a_n + (1-p) \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} (1-p)^j x_j,$$

then, with

$$\begin{aligned}
\sum_{j=k+1}^{n-1} \binom{n}{j} \binom{j}{k} (1-p)^j &= \frac{n!}{k!(n-k)!} \sum_{j=k+1}^{n-1} \frac{(n-k)!}{(n-j)!(j-k)!} (1-p)^j = \binom{n}{k} \sum_{j=k+1}^{n-1} \binom{n-k}{j-k} (1-p)^j \\
&= \binom{n}{k} (1-p)^k \sum_{j=1}^{n-k-1} \binom{n-k}{j} (1-p)^j \\
&= \binom{n}{k} (1-p)^k \Big((2-p)^{n-k} - 1 - (1-p)^{n-k} \Big)
\end{aligned}$$

we have

$$\begin{aligned}
x_n &= a_n + (1-p) \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} (1-p)^j \Big(a_j + (1-p) \sum_{k=0}^{j-1} \binom{j}{k} p^{j-k-1} (1-p)^k x_k \Big) \\
&= a_n + (1-p) \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} (1-p)^j a_j + (1-p)^2 \sum_{k=0}^{n-2} (1-p)^k p^{n-k-1} x_k \sum_{j=k+1}^{n-1} \binom{n}{j} \binom{j}{k} (1-p)^j \Big) \\
&=
\end{aligned}$$

We have

$$\begin{aligned}
w_{1,p} &= p, \\
w_{2,p} &= p^2 + 2p^2(1-p) = p^2(3-2p) = p(3p - 2p^2) < p = w_{1,p}, \\
w_{3,p} &= p^3 + 3p(1-p)(p w_{1,p} + (1-p) w_{2,p}) = p^3(1 + 3(1-p)(1 + (1-p)(3-2p))) \\
&\quad = p^3(1 + 3(1-p)(4 - 5p + 2p^2)) = p^3(13 - 27p - 21p^2 - 6p^3), \\
w_{3,p} &< p^3 + 3p(1-p)p = p^2(3-2p) = w_{2,p}, \\
w_{4,p} &= p^4 + 4p^3(1-p) w_{1,p} + 6p^2(1-p)^2 w_{2,p} + 4p(1-p)^3 w_{3,p}
\end{aligned}$$

$$
\begin{aligned}
w_{2,p} &= p^2 + 2p(1-p)w_{1,p} \\
w_{3,p} &= p^3 + 3p^2(1-p)w_{1,p} + 3p(1-p)^2 w_{2,p} \\
&\le p^3 + p^2(1-p)w_{1,p} + p(1-p)^2 w_{1,p} + 2p(1-p)w_{1,p} \\
&= (p + p(1-p) + (1-p)^2))p^2 + 2p(1-p)w_{1,p} = w_{2,p} \\
w_{4,p} &= p^4 + 4p^3(1-p)w_{1,p} + 6p^2(1-p)^2 w_{2,p} + 4p(1-p)^3 w_{3,p} \\
&\le p^4 + 4p^3(1-p)w_{1,p} + 3p^2(1-p)^2 w_{1,p} + p(1-p)^3 w_{2,p} + 3p(1-p)^2 w_{2,p} \\
&\le p^4 + p^3(1-p)w_{1,p} + p(1-p)^3 w_{2,p} + 3p^2(1-p)w_{1,p} + 3p(1-p)^2 w_{2,p} \\
&= p^2(p^2 + p^2(1-p) + (1-p)^3) - p(1-p)^3 p(1-2p)(1-p) + 3p^2(1-p)w_{1,p} + 3p(1-p)^2 w_{2,p} \\
w_{4,p} &= p^4 + 4p^3(1-p)w_{1,p} + 6p^2(1-p)^2 w_{2,p} + 4p(1-p)^3 w_{3,p} \\
&\le p^3(p + p(1-p) + (1-p)^2) + 3p^3(1-p)w_{1,p} + 3p^2(1-p)^2 w_{1,p}(p + p(1-p) + (1-p)^2)w_{1,p} + 3p(1-p)^2 w_{2,p} \\
&\le p^3 + 3p^2(1-p)w_{1,p} - 3p^2(1-p)^2 w_{1,p} + 5p^2(1-p)^2 w_{2,p} + 4p(1-p)^3 w_{2,p} \\
&= p^3 + p(1-p)w_{1,p}(3p^2 + 2p(1-p) + (1-p)^2) + 3p(1-p)^2 w_{2,p}
\end{aligned}
$$

We have $w_n - w_{n+1} = a_n - b_n$, where $a_0 = 1-p, b_0 = 0$,

$$
\begin{aligned}
a_n &= (1-p)\sum_{j=0}^{n-1}\binom{n}{j}p^{n-j}(1-p)^j a_j, \\
b_n &= p(1-p)^n w_n + (1-p)\sum_{j=1}^{n-1}\binom{n}{j}p^{n-j}(1-p)^j b_j
\end{aligned}
$$

For small p,

$$
\begin{aligned}
w_{1,p} &= p, \\
w_{2,p} &\approx 3p^2, \\
w_{3,p} &\approx 13p^3, \\
w_{4,p} &\approx 85p^4 \\
a_n &= \sum_{j=0}^{n-1}\binom{n}{j}a_j
\end{aligned}
$$

Is $n \mapsto w_{n,p}$ decreasing for some p?
Let us try to prove that $w_{n-1,p} - w_{n,p} \ge Cp(1-p)^{n-1}$ for some C.

$$
\begin{aligned}
w_0 - w_1 &= 1-p, \\
w_1 - w_2 &= -p(1-p)w_1 + p(1-p)(w_0 - w_1) = p(1-p)(w_0 - 2w_1), \\
w_2 &= p(1-(1-p)(1-2p)), \\
1 - w_2 &= w_0 - w_1 + w_1 - w_2 = (1 + p(1-2p))(w_0 - w_1) \\
w_2 - w_3 &= 3p(1-p)^2(w_1 - w_2), \\
w_2 - w_3 &= -p(1-p)^2 w_2 + p^2(1-p)(1-w_1) + 2p(1-p)^2(w_1 - w_2) \\
w_1 - w_2 &= p(1-3p+2p^2) = p(1-2p)(1-p) = p(1-2p)(w_0 - w_1), \\
w_2 - w_3 &= 3p(1-p)^2(w_1 - w_2), \\
w_3 &= w_2 - 3p(1-p)^2(w_1 - w_2), \\
w_3 - w_4 &= -p(1-p)^3 w_3 + (1-p)(p^3(w_0 - w_1) + 3p^2(1-p)(w_1 - w_2) + 3p(1-p)^2(w_2 - w_3))
\end{aligned}
$$

For $n = 2, 3, ...,$

$$
\begin{aligned}
w_{n-1,p} - w_{n,p} &= \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^j w_{j,p} - \sum_{j=0}^{n-1} \Big(\binom{n-1}{j} + \binom{n-1}{j-1} \Big) p^{n-j}(1-p)^j w_{j,p} \\
&= \Big(\sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{j+1} w_{j,p} \Big) - p(1-p)^{n-1} w_{n-1,p} \\
&\qquad\qquad - \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-j-1}(1-p)^{j+1} w_{j+1,p} \\
&= -p(1-p)^{n-1} w_{n-1,p} + \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{j+1}(w_{j,p} - w_{j+1,p}) \\
&= p(1-p)(\underbrace{p^{n-2}(1-p) - (1-p)^{n-2} w_{n-1}}_{=\sum_{j=1}^{n-2}(1-p)^j p^{n-1-j}(w_j - w_{j+1})}) + \sum_{j=1}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{j+1}(w_j - w_{j+1}) \\
&= -p^n + \sum_{k=0}^{n-2} p^{n-k}(1-p)^k w_k - p^{n-1-k}(1-p)^{k+1} w_{k+1} + \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{j+1}(w_{j,p} - w_{j+1,p}) \\
&\geq -p(1-p)^{n-1} w_{n-1,p} + Cp \sum_{j=1}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{2j} \\
&= -p(1-p)^{n-1} w_{n-1,p} + Cp(1-p+p^2)^{n-1} - Cp(1-p)^{2n-2}
\end{aligned}
$$

so $w_{n-1,p} > w_{n,p}$ if $w_{n-1,p} < \frac{p^{n-2}}{(1-p)^{n-3}}$.

1 A version of the game in continuous time

Consider a Poisson process at rate x, as well as $X_1, ..., X_n \sim \exp(1)$, which are independent. Then, we are asking for the probability that no two points T_k, T_{k+1} of the x-Poisson process happen with $X_1, ..., X_n \notin \{T_k, T_{k+1}\}$ and $X_i > T_{k+1}$ for at least one i. Again, call this probability w_n. We obtain the recursion, with $w_0 = 1$,

$$w_n = \sum_{k=0}^{n-1} \frac{n \cdots (k+1) \cdot x}{(n+x) \cdots (k+1+x)(k+x)} w_k = x \sum_{k=0}^{n-1} \frac{\Gamma(n+1)\Gamma(k+1+x)}{\Gamma(k+2)\Gamma(n+1+x)} w_k.$$

Note that for $x = 1$,

$$w_n = \frac{1}{n+1} \sum_{k=0}^{n-1} w_k = \frac{1}{n+1} \Big(1 + \sum_{k=1}^{n-1} w_k\Big),$$

which is solved by $w_1 = w_2 = ... = \frac{1}{2}$. Recursively, using the recursion for w_{n-1} in the fourth equality,

$$\begin{aligned}
w_n - w_{n-1} &= \sum_{k=0}^{n-1} \frac{n \cdots (k+1) \cdot x}{(n+x) \cdots (k+1+x)(k+x)} w_k - \sum_{k=0}^{n-2} \frac{(n-1) \cdots (k+1) \cdot x}{(n-1+x) \cdots (k+1+x)(k+x)} w_k \\
&= \frac{nx}{(n+x)(n-1+x)} w_{n-1} + \sum_{k=0}^{n-2} \Big(\frac{n \cdots (k+1) \cdot x - (n+x)(n-1) \cdots (k+1) \cdot x}{(n+x) \cdots (k+1+x)(k+x)} w_k \\
&= \frac{nx}{(n+x)(n-1+x)} w_{n-1} - x^2 \sum_{k=0}^{n-2} \frac{(n-1) \cdots (k+1)}{(n+x) \cdots (k+1+x)(k+x)} w_k \\
&= \frac{nx}{(n+x)(n-1+x)} w_{n-1} - \frac{x}{n+x} w_{n-1} = \frac{x}{n+x} \Big(\frac{n}{n-1+x} - 1 \Big) w_{n-1}.
\end{aligned}$$

From the last line, it is clear that

$$w_n - w_{n-1} < 0 \quad \iff \quad x > 1.$$

In particular, $n \mapsto w_n$ is decreasing for $x > 1$.